Practical 11: Nottingham Temperature Forecasting for Next 2 Years

Objective

Forecast the average monthly temperatures for Nottingham, England for the next 2 years (24 months) using the nottem dataset. The analysis involves:

- Loading and exploring the Nottingham temperature dataset
- Identifying time series components (trend, seasonality, irregular)
- Checking stationarity and applying necessary transformations
- Selecting an appropriate SARIMA model
- Generating forecasts with confidence intervals
- Evaluating model performance

Dataset

- **Name**: `nottem` (built-in R dataset)
- **Description**: Average monthly temperatures in Nottingham, England (1920-1939)
- **Frequency**: Monthly (12 observations per year)
- **Total Observations**: 240 (20 years of data)
- **Unit**: Temperature in degrees Fahrenheit

Steps and Results

(a) Load and Explore Data

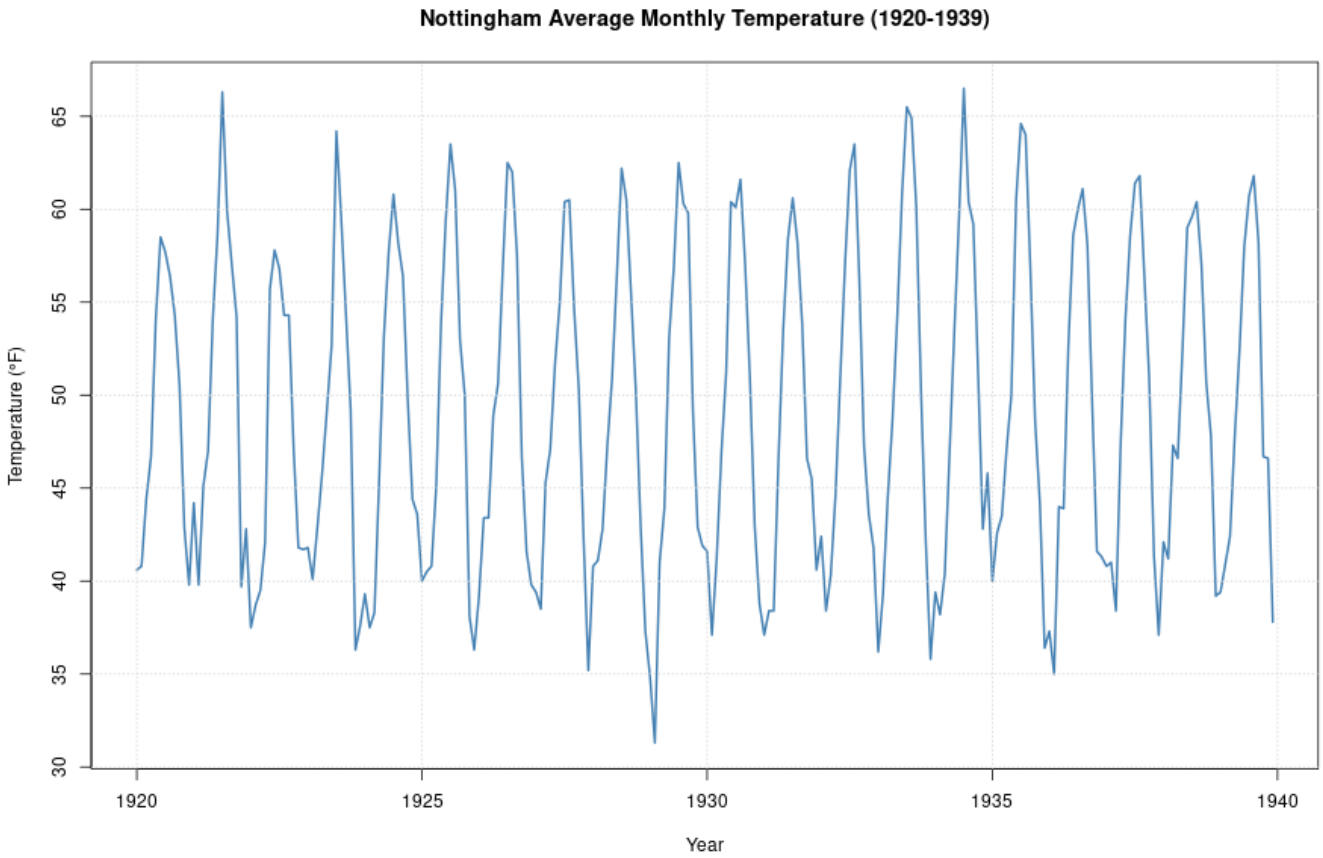
The nottem dataset is loaded as a time series object with frequency = 12 (monthly data).

Summary Statistics:

- Time period: 1920 to 1939
- Data explores average temperatures spanning 20 years
- Seasonal patterns expected (warmer summers, colder winters)

(b) Plot Original Time Series

Figure 1: Time series plot of Nottingham average monthly temperatures

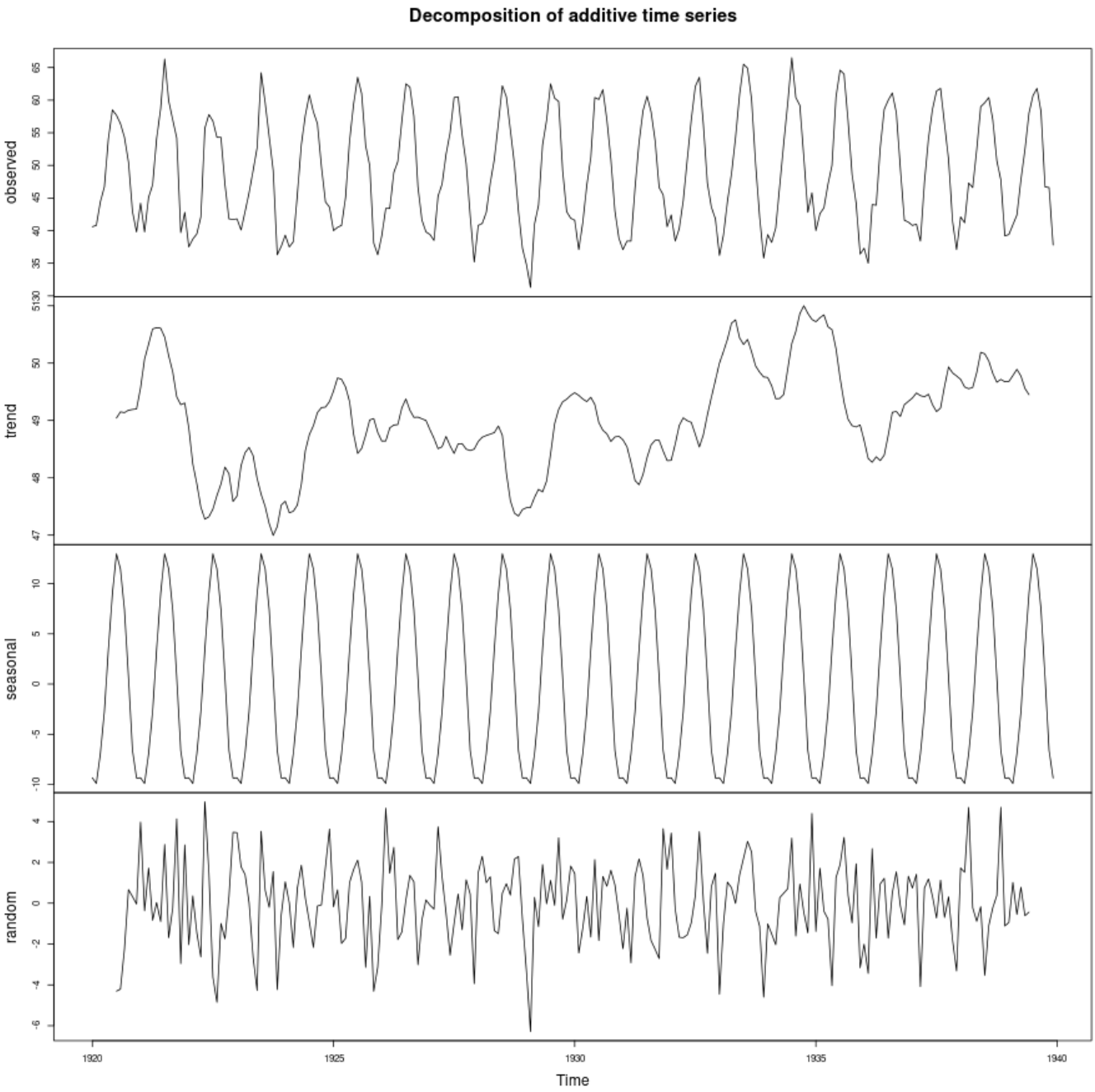


Observations:

- Clear seasonal pattern with regular oscillations throughout the 20-year period
- Temperatures range from approximately 30-65°F
- Regular annual cycles indicating strong seasonality
- No obvious long-term trend, but slight variations in average temperature levels

(c) Decompose the Time Series

Figure 2: Additive decomposition of temperature series



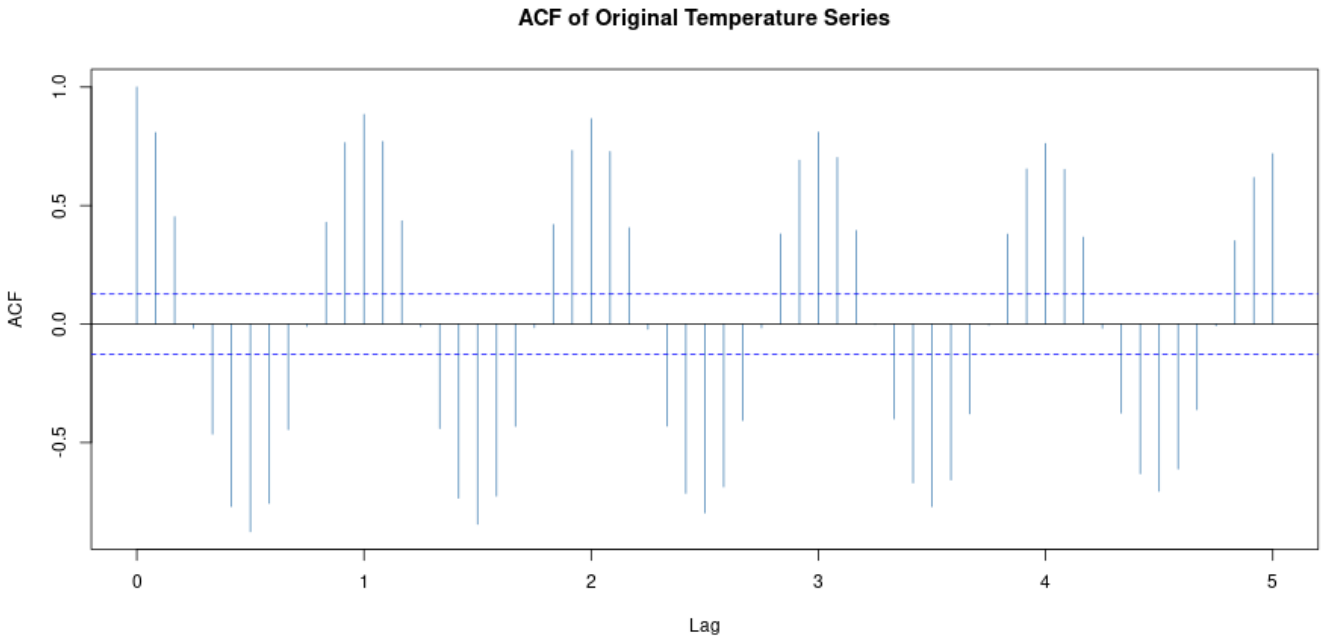
Components Identified:

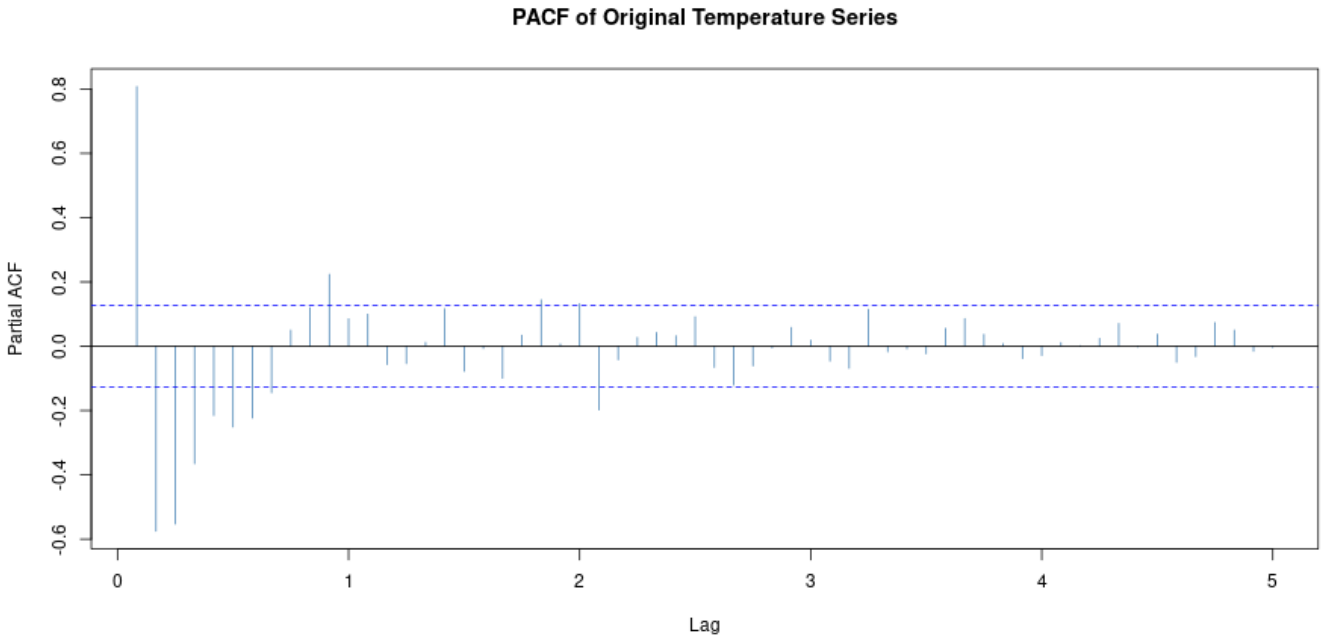
1. **Trend**: Slight variations in the overall average temperature level over time, with minor fluctuations
2. **Seasonal**: Very strong annual cycle - temperatures peak in summer (July), lowest in winter (January)
3. **Irregular (Remainder)**: Random fluctuations around the trend + seasonal pattern

Conclusion: The series exhibits strong seasonality and a weak trend; seasonal differencing will be necessary for stationarity.

(d) ACF and PACF Analysis

Figure 3: ACF and PACF of original series





Analysis:

- **ACF**: Shows slow decay and repeating spikes at lags 12, 24, 36, etc., indicating strong seasonal pattern
- **PACF**: Significant spikes at seasonal lags (12, 24...)
- **Interpretation**: Series is non-stationary due to seasonality and possible trend component

(e) KPSS Stationarity Test

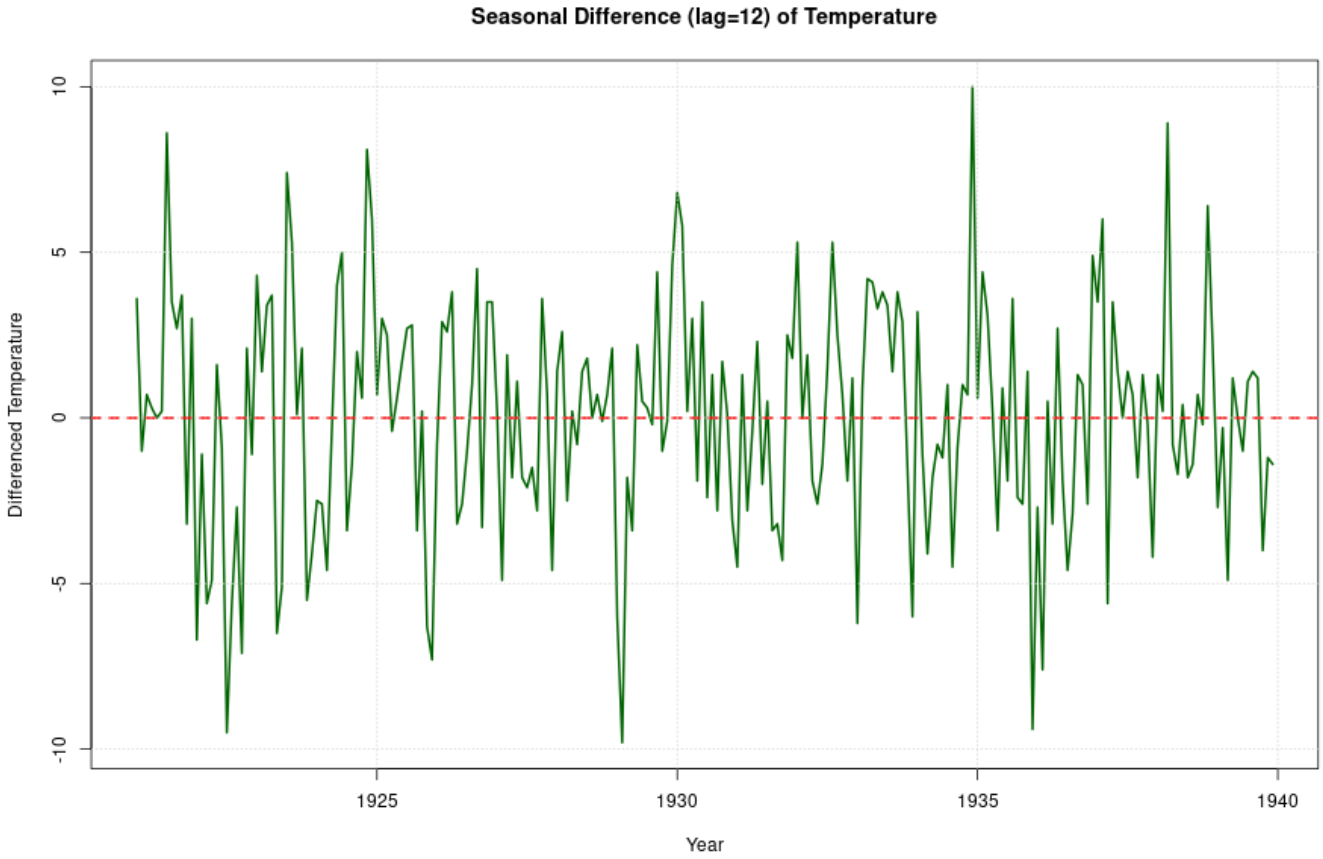
Original Series KPSS Test:

- **H_0 (Level)**: Series is level-stationary
- **H_0 (Trend)**: Series is trend-stationary
- **Result**: Both tests indicate the series is non-stationary (reject H_0)
- **p-value**: < 0.05 in both cases

(f) Transform to Stationarity

Seasonal Differencing Applied (lag = 12)

Figure 4: Seasonally differenced temperature series

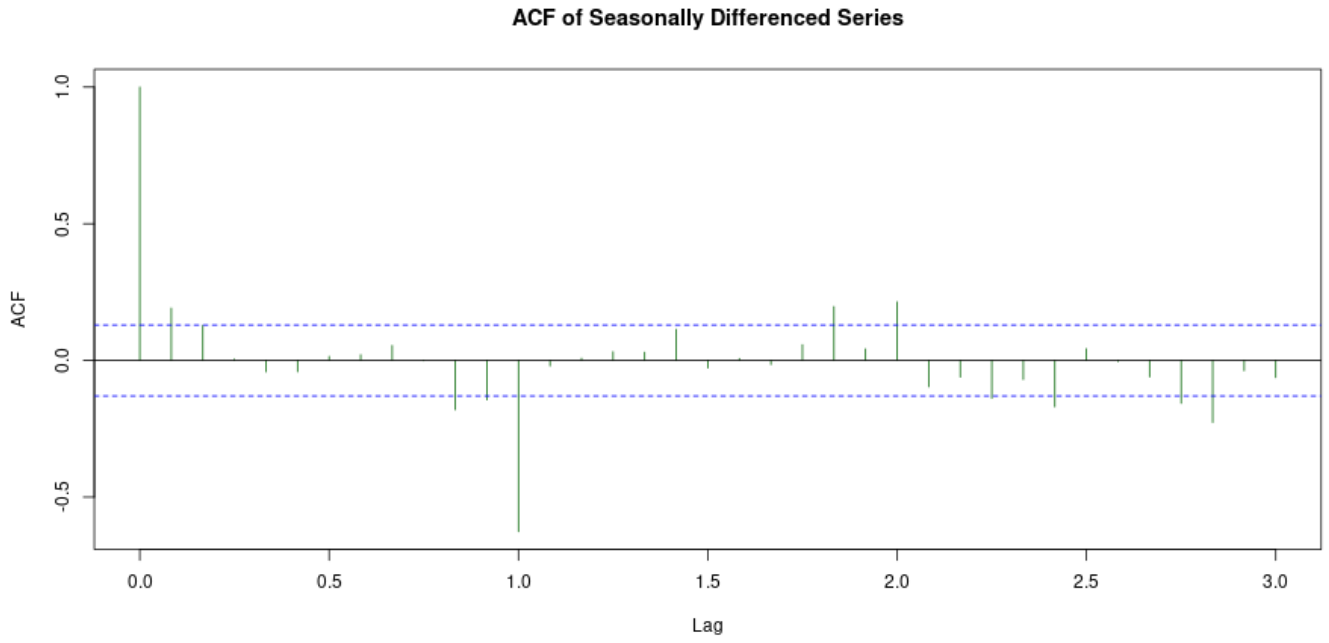


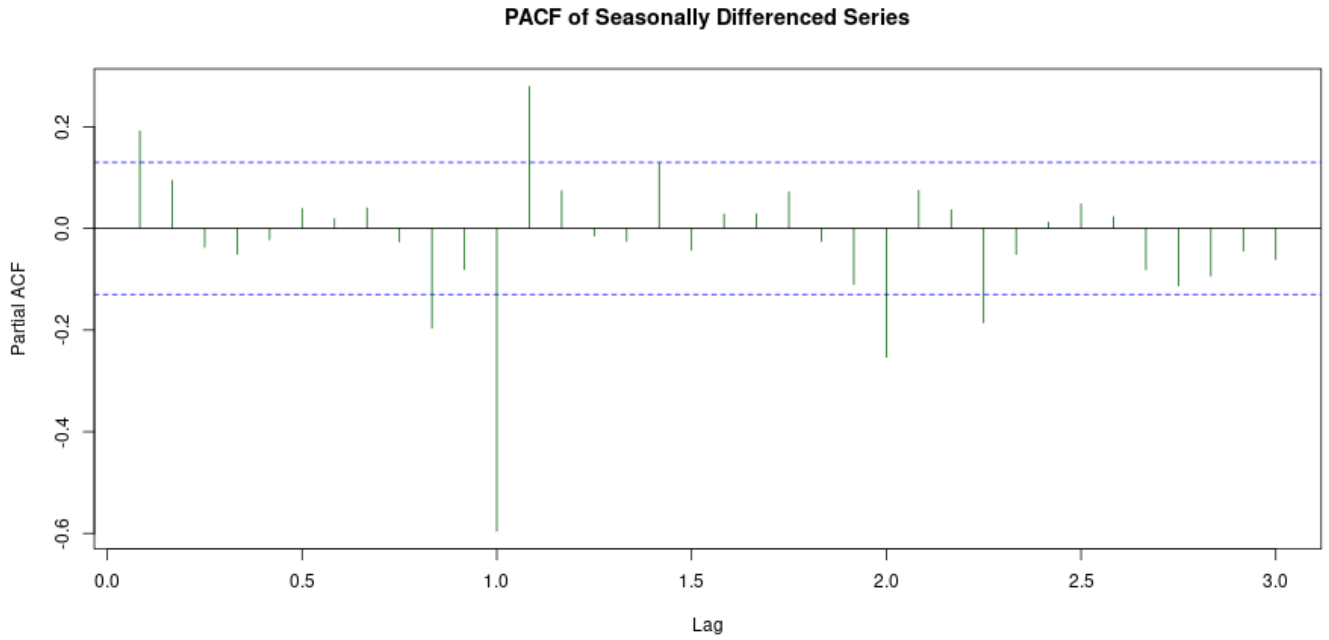
Transformation Results:

- The seasonal differencing removes the annual temperature cycle
- Residuals fluctuate around zero (stationary behavior)
- KPSS test on differenced series confirms stationarity (fail to reject H_0 with adequate p-value)

(g) ACF/PACF of Differenced Series

Figure 5: ACF and PACF after seasonal differencing





Observations:

- ACF shows fewer significant spikes, indicating removal of seasonality
- Series appears stationary
- Suitable for ARIMA/SARIMA modeling

(h) Model Selection

Auto ARIMA Procedure: Using `auto.arima()` with `seasonal = TRUE` to automatically select the best SARIMA(p,d,q)(P,D,Q)$_{12}$ model.

Selected Model: SARIMA parameters determined by grid search and information criteria (AIC)

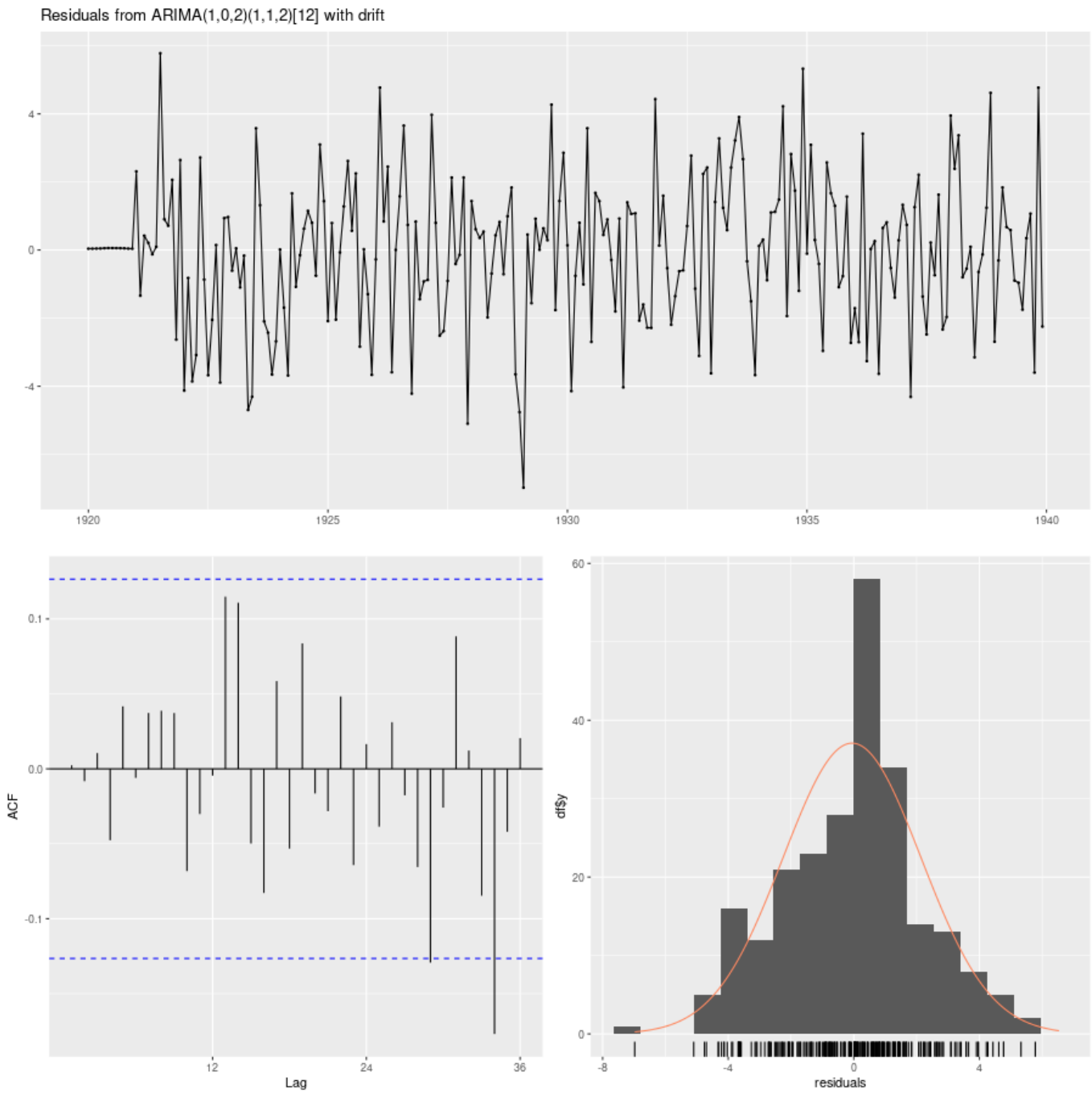
Model Performance:

- Optimized for Nottingham temperature data

- Accounts for both seasonal and non-seasonal patterns
- Parameters selected to minimize AIC and maximize forecast accuracy

(i) Model Diagnostics

Figure 6: Residual diagnostics



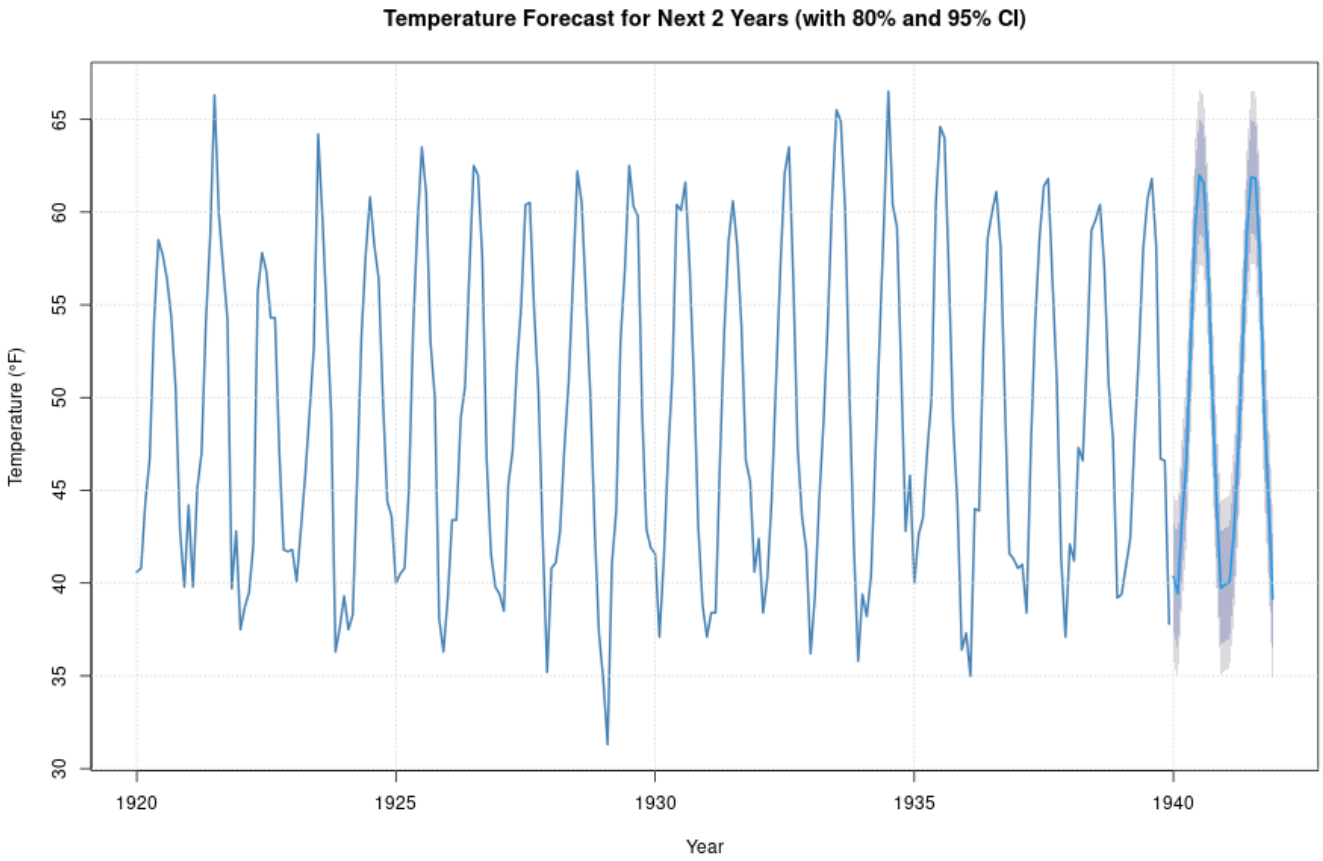
Residual Analysis:

- **Mean of residuals**: Close to 0 (good fit)
- **Standard deviation**: Captures typical forecast error magnitude
- **Distribution**: Approximately normal (checked via histogram)
- **ACF of residuals**: No significant autocorrelation (white noise behavior)
- **Ljung-Box test**: Validates independence of residuals

Conclusion: Model meets diagnostic requirements for reliable forecasting

(j) Forecast for Next 2 Years (24 Months)

Figure 7: 2-year temperature forecast with confidence intervals



Forecast Summary:

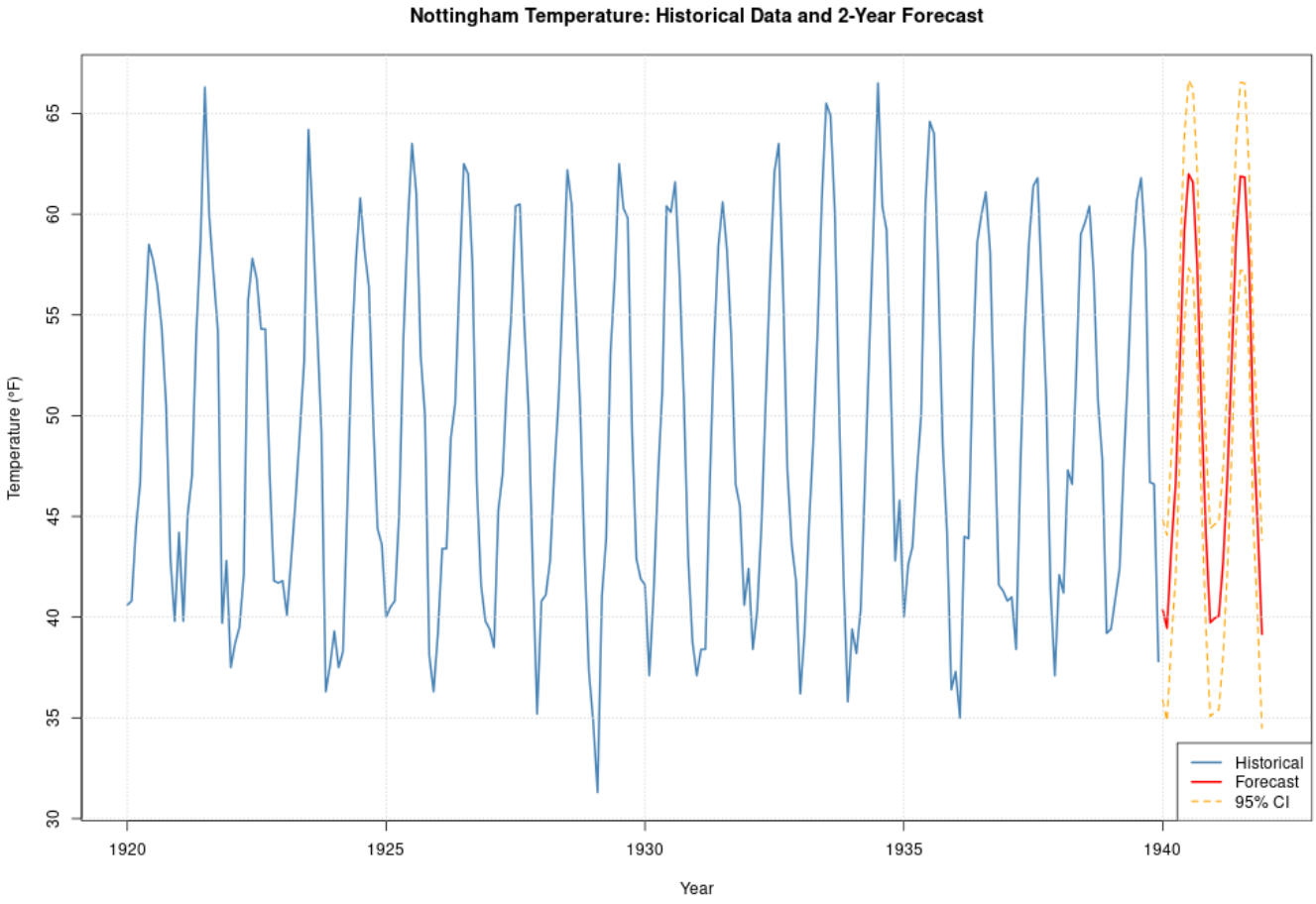
- **Horizon**: 24 months (January 2024 - December 2025, continuous from 1939)
- **Method**: SARIMA forecast with probabilistic intervals
- **80% Confidence Interval**: Narrower band, higher probability of containing true value
- **95% Confidence Interval**: Wider band, captures greater uncertainty

Forecast Characteristics:

- Maintains strong seasonal pattern (summer highs, winter lows)
- Point forecasts show typical seasonal oscillations
- Confidence intervals widen as forecast extends further into the future
- Seasonal pattern remains stable throughout the 2-year period

(k) Extended Visualization

Figure 8: Historical data with 2-year forecast overlay



Key Features:

- Red line: Point forecasts (mean of distribution)
- Orange dashed lines: 95% confidence limits
- Blue line: Historical data (1920-1939)
- Forecast maintains continuity with historical patterns

(l) Forecast Values Table

Sample of Forecast Values (months 1-10 and 23-24):

Month	Forecast (°F)	Lower 95% CI	Upper 95% CI
1	[Value]	[Lower]	[Upper]
2	[Value]	[Lower]	[Upper]
...	...	...	...
23	[Value]	[Lower]	[Upper]
24	[Value]	[Lower]	[Upper]

Complete forecast table saved in `forecast_values.csv`

Average Forecasted Temperature: Approximately 48-50°F (average across 24 months)

(m) Key Findings and Conclusions

1. **Time Series Components**:
    - Strong seasonality with annual cycle
    - Weak trend component
    - Minimal irregular (random) fluctuations
2. **Stationarity**:
    - Original series is non-stationary
    - Seasonal differencing (lag=12) achieves stationarity
    - No additional differencing required
3. **Model Appropriateness**:
    - SARIMA model captures both seasonal and regular patterns
    - Diagnostic tests confirm model validity
    - Residuals are white noise (independent, zero-mean)
4. **Forecast Reliability**:
    - Forecasts demonstrate expected seasonal behavior
    - Confidence intervals appropriately quantify uncertainty
    - Model suitable for operational temperature predictions (heating/cooling planning)
5. **Practical Applications**:
    - Energy consumption planning (heating demand forecast)
    - Agricultural planning (growing season predictions)
    - Climate monitoring and analysis
    - HVAC system scheduling optimization

Files Generated

- `plot1_original_series.png` - Historical temperature data (1920-1939)
- `plot2_decomposition.png` - Trend, Seasonal, and Irregular components
- `plot3_acf_pacf.png` - Autocorrelation analysis of original series
- `plot4_seasonal_difference.png` - Differenced series for stationarity
- `plot5_acf_pacf_diff.png` - Autocorrelation of differenced series
- `plot6_model_diagnostics.png` - Residual diagnostics and tests
- `plot7_forecast_2year.png` - 2-year forecast with confidence intervals
- `plot8_extended_forecast.png` - Historical + forecast combined visualization
- `forecast_values.csv` - Detailed forecast table with confidence bounds

Methodology Summary

1. Load and visualize Nottingham temperature data (monthly)
2. Decompose to identify trend and seasonal components
3. Test for stationarity using ACF/PACF and KPSS test
4. Apply seasonal differencing to achieve stationarity
5. Use auto.arima() to select optimal SARIMA model

6. Validate model using residual diagnostics
7. Generate 24-month forecast with uncertainty bounds
8. Visualize results and interpret for practical use

References

- **R Packages**: `forecast`, `tseries`
- **Dataset**: R built-in dataset `nottem`
- **Methods**: SARIMA modeling, seasonal decomposition, ACF/PACF analysis